



REGISTERED REPORT

How retributive motives shape the emergence of third-party punishment across intergroup contexts

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Abstract

This study examines how retributive motives—the desire to punish for the purpose of inflicting harm in the absence of future benefits—shape third-party punishment behavior across intergroup contexts. Six- to nine-year-olds ($N=151$, $M_{\text{age}}=8.00$, $SD_{\text{age}}=1.15$; 54% White, 18% mixed ethnicities, 17% Asian American; 46% female; from the USA) could punish ingroup, outgroup, or non-group transgressors by removing positive resources and allocating negative ones. Both punishments were described as retributive, yet allocating negative resources was perceived as more retributive than removing positive ones. We predicted that children would punish outgroups more so than ingroups and that this effect would be especially pronounced when punishment is perceived as particularly retributive. The results did not align with this prediction; instead, children similarly punished all agents.

INTRODUCTION

A striking feature of human societies is that we enforce norms even when we merely witness a transgression, so-called *third-party punishment* (e.g., Fehr & Fischbacher, 2004). For example, we may publicly shame a colleague for stealing someone's lunch from a shared refrigerator or send a child to timeout for bullying their sibling. In each of these cases, the punisher is not directly affected by the transgression but still pursues punishment of the wrongdoer, sometimes even at a great cost. Ample experimental research has documented this behavior in both children (e.g., McAuliffe et al., 2015; Yudkin et al., 2020) and adults (e.g., Fehr & Fischbacher, 2004) and has found that it emerges across diverse cultures from childhood into adulthood (Henrich et al., 2010; House et al., 2020).

The work on the emergence of third-party punishment behavior has principally focused on individuals' willingness to punish strangers. Despite this focus, punishment in everyday life rarely occurs only among unknown individuals and, instead, occurs among people within specific relational contexts. One relevant relational consideration is identity, such as race, gender,

or nationality. Whether a bystander shares or does not share an identity with a transgressor may influence whether the bystander is willing to pursue punishment. Indeed, individuals organize themselves around *group membership* (whether they are in the same or different group from another person), and research has found that intergroup context shapes our social decision-making (e.g., Baron & Dunham, 2015; Dunham, 2018; Dunham et al., 2011) and our reasoning about others' behaviors (e.g., DeJesus et al., 2014; Rhodes & Chalik, 2013) in profound ways, even in childhood. With this past work in mind, the present study aims to better understand how third-party punishment behavior emerges across different intergroup contexts, with a particular emphasis on understanding *why* we punish outgroup transgressors.

The role of intergroup context in third-party punishment

To gain traction on the question of how intergroup context modulates punitive behavior, it is useful to first turn to prevailing evolutionary discussions of why—at

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the ultimate level—we engage in third-party punishment. The prominent view in the literature is that third-party punishment serves as an important mechanism (among many) to promote cooperation within groups (Balliet et al., 2011; Boyd & Richerson, 1992; Henrich & Muthukrishna, 2021; Mathew & Boyd, 2011). Supporting this proposal, some lines of research find that an individual's willingness to punish transgressors is sensitive to whether punishment has the potential to precipitate positive societal outcomes, such as deterring bad behavior or establishing a norm (so-called *consequentialist motives*; Bentham, 2000). For example, children and adults punish more when doing so can deter bad behavior compared to when it cannot (Crockett et al., 2014; Funk et al., 2014; Marshall et al., 2021; Sarin et al., 2021; Twardawski & Hilbig, 2020) and also judge punishment as having the potential to deter bad behavior in the future (Bregant et al., 2016; Dunlea & Heiphetz, 2021; Marshall et al., 2022).

When it comes to the intergroup context, the position that punishment maintains cooperation within groups generates a clear prediction: if punishment, at the ultimate level, helps to sustain cooperation, it stands to reason that third-party punishment should be biased toward those with whom the punisher is most likely to interact, namely in-group members. Indeed, Shinada et al. (2004) articulate this view clearly: “If the goal of punishment is to improve the behavior of potential free riders, then punishment should be directed to potential free riders in the punisher's own group, where the benefits of cooperation are shared among group members, including the punisher” (p. 381). Based on these discussions, psychological researchers have described this position as the “*norm-focused hypothesis*” (McAuliffe & Dunham, 2016): third-party punishment evolved as a mechanism to sustain cooperation within groups. Indeed, researchers have found evidence consistent with this hypothesis in adults (e.g., Shinada et al., 2004) and children (e.g., Gonzalez-Gadea et al., 2022; Leshin et al., 2022).

Despite this theoretical background and empirical findings, however, a preponderance of other studies with adults and children find different results. In particular, a sizable amount of research finds that adults and children actually punish outgroup members *more* than ingroup members (Baumgartner et al., 2012; Chapman et al., 2020; Delton & Krasnow, 2017; Gummerum et al., 2009; Guo et al., 2020; Jordan et al., 2014; Lieberman & Linke, 2007; Schiller et al., 2014). Complicating matters further, other work finds that children and adults punish ingroup and outgroup transgressors similarly under some circumstances (Bernhard et al., 2006; Gummerum et al., 2009; Schmidt et al., 2012; Schuhmacher & Kärtner, 2019; Yudkin et al., 2020). These results then raise the question of why these contrasting patterns of results emerge in both developmental and adult samples, despite theoretical reasons to expect greater punishment of ingroup transgressors over outgroup ones.

One way to gain insight into why these contrasting findings emerge is to first focus on why we punish outgroup transgressors—the question at the core of the current research. Unlike the evolutionary discussions surrounding why we punish ingroup transgressors, conversations surrounding why individuals punish outgroup transgressors are less well-formalized in the literature. Efforts to address the question of outgroup punishment largely draw on discussions from the intergroup bias literature more generally (e.g., Hewstone et al., 2002). That is, researchers have argued that the tendency to punish outgroup transgressors more than ingroup transgressors may be a byproduct of more general affective preferences in favor of ingroup members and against outgroup members—what has been called the “*mere preferences hypothesis*” (McAuliffe & Dunham, 2016; see Krasnow et al., 2016, for an alternative explanation). For example, we may want to avoid punishing ingroup transgressors because we generally like them and do not want to harm them via punishment (i.e., “ingroup love”; ingroup favoritism). At the same time, we may also want to actively punish outgroup transgressors because we generally dislike them and want to harm them via punishment (i.e., “outgroup hate”, outgroup discrimination).

Though these connections have yet to be explicitly drawn out, the “mere preferences hypothesis” has important links to philosophical discussions about so-called *retributive motives* (i.e., “just deserts”; Kant, 2011): motives that are not contingent on the outcome of punishment and are instead tethered to the desire to inflict deserved suffering (“just deserts”). These discussions about retribution typically argue that punishment is inherently retributive because all forms of punishment involve inflicting some sort of harm on a transgressor. Importantly, though, the extent to which a punishment is retributive may vary. For instance, fining someone US\$100 rather than US\$1 for a crime is straightforwardly more retributive because a fine of US\$100 inflicts greater harm than a fine of US\$1. Even further, different types of punishment also vary in their retributive value. For example, making someone do hard manual labor may be perceived as more retributive than fining someone US\$100.

Research has found that children and adults are motivated to punish because of retributive concerns; that is, children and adults punish even when doing so cannot possibly precipitate prosocial outcomes (e.g., Crockett et al., 2014; Lee et al., 2024; Marshall et al., 2021; Twardawski & Hilbig, 2020). This research, however, has not examined how retributive motives may function in intergroup contexts. Relating the psychological discussions about intergroup bias to the philosophical ones about retributive motives, it is possible—as articulated above—that individuals punish outgroup transgressors more than ingroup transgressors because individuals want to avoid inflicting retribution upon ingroup members, want to inflict retribution on

outgroup members, or a combination of these. Indeed, methodological features of past studies provide some support for these possibilities. Most third-party punishment studies finding greater outgroup punishment (e.g., Jordan et al., 2014) involve making decisions in settings particularly likely to invoke retributive motives, such as contexts involving anonymous others where future interaction is very unlikely. Because these features (i.e., anonymity and lack of future interaction) are not systematically manipulated in experimental studies, we do not know whether participants' greater willingness to punish outgroup members can be explained by them. With this background in mind, the present study aims to help clarify how intergroup context shapes third-party punishment, with a particular emphasis on the role of retributive motives in outgroup punishment.

Why development?

The present work uses a developmental approach to examine punishment in its earlier form, at age six, and its expression in late childhood, by age nine. By using this approach, we can gain insight into how retributive motives shape third-party punishment in intergroup contexts during a developmental window in which the tendency to punish strengthens (e.g., House et al., 2020; McAuliffe et al., 2015).

We aim to test children starting at the age of six because research has demonstrated that children at this age exhibit a consistent willingness to punish transgressors in response to many different types of antisocial actions. For example, children punish fairness violations (e.g., McAuliffe et al., 2015; USA), theft (e.g., Riedl et al., 2015; Germany), and property destruction (e.g., Marshall et al., 2021; USA). Children by age six also engage in different forms of punishment, including the removal of positive resources (e.g., Yudkin et al., 2020; USA) and the allocation of negative ones (e.g., Kenward & Öst, 2015; Sweden). Moreover, by age six, we know that both “ingroup love” and “outgroup hate” shape children's social behavior (Buttelmann & Böhm, 2014) and that children's third-party punishment behavior is reliably motivated by retribution (Marshall et al., 2021; McAuliffe & Dunham, 2021) and influenced by intergroup dynamics (e.g., Jordan et al., 2014; Yang et al., 2021). In sum, by testing children at this age, we can gain insight into whether third-party punishment behavior is shaped by intergroup dynamics around the time in which children are shown to consistently engage in such behavior across a variety of situations and contexts.

Beyond 6 years of age, we are also interested in studying the expression of third-party punishment through middle childhood, up until age nine. We wanted to do so because previous work suggests that it is roughly during this developmental window that punishment

has emerged across contexts, as discussed above, and grows in strength (e.g., House et al., 2020; McAuliffe et al., 2015). Indeed, older children tend to exhibit a greater willingness to punish relative to younger children, illustrating that children's motivation to engage in punishment behavior during this period in development becomes stronger.

In addition to theoretical reasons for testing how intergroup bias and retributive motives shape third-party punishment in children, there are also methodological reasons to do so. Younger children are typically more willing to reveal their explicit intergroup bias than older children and adults (e.g., Raabe & Beelmann, 2011). Within the domain of punishment and intergroup bias, there is some support for this possibility in adult research. In particular, adults will punish ingroup and outgroup members equally unless they are thinking more intuitively, in which case, they will punish outgroup members more than ingroup members (Yudkin et al., 2016). Similarly, adults will engage in retributive behavior, yet they explicitly reject the notion that their punishment decisions are motivated by such concerns. This finding suggests that adults outwardly reject retribution yet will engage in it anyway (e.g., Carlsmith, 2008; Crockett et al., 2014). It is thus possible that adults' motivations to punish are, at least in part, influenced by how they think they *should* punish and not actually by what they desire to do. By testing children, we can potentially generate a clearer picture of intergroup bias in punishment.

The current study's methodological approach and aims

The present study specifically examines how retributive motives shape the emergence of third-party punishment across intergroup contexts, with a particular theoretical focus on the motives underlying outgroup punishment. To address this, we will investigate whether children between the ages of six and nine punish outgroup transgressors more than ingroup ones in explicitly retributive contexts and whether this bias is especially pronounced when punishment is perceived as particularly retributive. Below, we outline our primary research aims in addition to our exploratory aims and connect these to our methodological approach.

Primary aims

The present work is designed to address two primary aims, which are outlined in Table 1. To address these aims, we developed a three-part procedure that allows us to examine children's third-party retributive punishment in intergroup contexts. First, children are assigned to a group using a previously established paradigm (i.e., a “maximal” group; Yang & Dunham, 2019). Second, participants hear

TABLE 1 Proposed aims with associated research questions.

Research questions	
Primary Aim 1	Do children punish outgroup transgressors more than ingroup transgressors in explicitly retributive contexts?
Primary Aim 2	Do children engage in <i>even greater</i> outgroup punishment in contexts where punishment is perceived as especially retributive (i.e., Negative Resource Allocation)?
Exploratory Aim 1	In what ways are the hypothesized patterns dependent on age?
Exploratory Aim 2	How does group context change evaluations of and emotional responses to antisocial behavior? Do potential differences predict variation in punishment behavior?

six stories that vary on two dimensions, both of which are manipulated within-subjects: (a) whether the main character acts antisocially or neutrally, and (b) whether the main character is portrayed as an ingroup, outgroup, or non-group member compared to the participant. Third, participants can punish the main character in two non-mutually exclusive ways: they can remove any number of valued resources (good sounds) or allocate any number of aversive resources (bad sounds). In classical psychological literature (Skinner, 1953), the former type of punishment has been referred to as “negative punishment” and the latter has been referred to as “positive punishment.” Because these terms could potentially create confusion within the context of our experimental design, we will instead refer to the former type of punishment as “Positive Resource Removal” and the latter as “Negative Resource Allocation” (Marshall & McAuliffe, 2022), where “positive” and “negative” refer to the *kinds* of resources (i.e., good and bad sounds) being removed or allocated. In line with past research (Marshall et al., 2021), we will instruct participants that both forms of punishment are retributive because neither type of punishment will be delivered alongside any sort of message thereby removing the possibility that the transgressors could learn a lesson as a result of punishment.

Two key methodological aspects of the punishment task allow us to address our two primary aims. With respect to Primary Aim 1, both forms of punishment are described as retributive, so we can assess whether children punish outgroup transgressors more in such contexts (*Primary Aim 1*; Table 1). Such a finding would conceptually replicate previous findings documenting greater outgroup punishment in children (e.g., Jordan et al., 2014) and also provide insight into why children are interested in punishing outgroup members. With respect to Primary Aim 2, we anticipate that children will view one form of punishment, particularly Negative Resource Allocation, as more harmful than another form of punishment, particularly Positive Resource Removal. Given this, we can examine whether children are particularly likely to punish outgroup members in contexts where punishment is especially retributive.

Beyond allowing us to determine whether retributive motives shape a willingness to engage in outgroup punishment, including these two ways of punishment is also important for methodological reasons. Past developmental research on punishment has, for the most part, constrained participants' options to either punishing or not punishing at all (e.g., Marshall et al., 2021; McAuliffe et al., 2015); studies rarely provide a variety of options to participants who want to respond to an antisocial action (although see McAuliffe & Dunham, 2021; Yang et al., 2021). The present study, because it measures punishment continuously and because it provides two ways of punishment, reduces the “experiment demand” on participants to punish in any particular way and gives participants flexibility in their own punitive decisions. Had we only provided participants with one way to punish, then their decisions could—at least in part—reflect the constrained option set and not their true interests. Further, because we include both Negative Resource Allocation punishment and Positive Resource Removal punishment, our study brings together two previously distinct punitive paradigms that have not yet been integrated into a single study design.

Exploratory aims

Beyond our two Primary Aims, we will also investigate two exploratory research questions. First, do the hypothesized patterns of punishment vary with age (*Exploratory Aim 1*; Table 1)? Note that we do not expect age interactions because previous research on intergroup bias in punishment has not found any such effects in a sample of similarly-aged children (e.g., Jordan et al., 2014). However, we expect to see a main effect of age, such that punishment increases over time (e.g., House et al., 2020). Second, how does group context change evaluations of and emotional responses to antisocial behavior (*Exploratory Aim 2*; Table 1)? To test this, we will ask participants to report whether they think the main character's behavior is good or bad (e.g., Chapman et al., 2020; Schuhmacher & Kärtner, 2019), as well as how angry, sad, or happy participants feel in response to the main character. This feature of the design can provide insight into the evaluative and emotional mechanisms that may explain intergroup bias in punishment. For example, it is possible that greater anger (but not sadness) in response to outgroup antisocial actions promotes greater retributive behavior toward such actors. In support of this, ample research has linked anger (but not sadness) with third-party punishment behavior in adults (e.g., Gummerum et al., 2016) and preliminary work has shown similar effects in older children (9-year-olds; Gummerum et al., 2020, 2022; see Marshall & McAuliffe, 2024, for a review) but not in younger children (5-year-olds; Marshall et al., 2021).

In summary, the present study will assess how retributive motives shape third-party punishment across

intergroup contexts by addressing two primary research aims. First, the present study can establish whether children punish outgroup members more than ingroup members in explicitly retributive contexts. Second, the current work also investigates the possibility that intergroup bias in punishment is even more pronounced in contexts where punishment is perceived as particularly retributive (e.g., Negative Resource Allocation). By addressing these two aims, the present work is well-positioned to help clarify our understanding of how retributive motives shape punitive behavior across intergroup contexts and provide a theoretical and empirical foundation for future investigations into questions about punishment and cooperation in development.

METHOD

Participants

We tested a total of 151 participants, which cohered with our goal of 148 participants (see [Supporting Information](#) for full information about planned sample size and power analyses), online through the Zoom platform. An additional three participants were tested but excluded because they had already participated ($n=2$) or their parent or guardian disclosed a disability ($n=1$). Of the 151 final participants ($M_{\text{age}}=8.00$, $SD_{\text{age}}=1.15$), 35 were 6-year-olds, 39 were 7-year-olds, 38 were 8-year-olds, and 39 were 9-year-olds. Seventy participants identified as female and 81 identified as male. Twenty-five participants identified as Asian or Asian American, 5 identified as Black or African American, 6 identified as Hispanic or Latino, 1 identified as Middle Eastern, 1 identified as Native American and 81 identified as White or Caucasian; the remaining participants indicated mixed racial or ethnic backgrounds ($n=27$) or did not report their ethnicity ($n=5$). All study sessions were conducted in English and with an experimenter. Four children did not pass a comprehension check, although we still included these participants in the final sample (as specified in our analysis plan). All participants were recruited through a lab-based database that was generated through Facebook ads that targeted parents in the United States. Additional demographic information was collected regarding our participants, as it is standard practice in our lab for parents to fill out a demographic form alongside the consent; because these data are not theoretically relevant, those data can be found on Open Science Framework (OSF): osf.io/6uxpe.

Study procedures

The study procedure involved three components: (1) Group Induction, (2) “Sound Box” Task Explanation, and (3) Test Trials. For ease of understanding, all

verbatim scripts and a demonstration of the paradigm are on OSF. The study was pre-registered on OSF on March 1st, 2022 prior to beginning testing on March 4, 2022 at the following link: <https://doi.org/10.17605/OSF.IO/ZGTNV>. Testing ended in July 2022. We adhered to our Stage-1 manuscript procedures with one exception; we added some final questions at the end of our procedure, which are described in detail in the [Supporting Information](#).

Group induction

Children were assigned to groups using a “maximal group” manipulation from previous research by Yang and Dunham (2019); see [Figure 1](#) for full information. First, the experimenter introduced participants to two groups: the green group and the blue group. Second, the experimenter introduced participants to a “machine” (an online spinner wheel). The experimenter instructed participants to look into their video camera while the experimenter spun the wheel, revealing the participant's group membership. In actuality, whether participants were assigned to the blue or green group was counter-balanced. Next, the participants selected an avatar to represent themselves during the rest of the experiment. Finally, we asked several manipulation check questions to ensure that we had successfully induced ingroup bias. Specifically, participants were asked a total of six partner choice questions—three questions about a class project partner (“If you were working on a class project, who would you want to work with?”) and three questions about a play partner (“If you wanted to play on the seesaw, who would you want to play with?”).

“Sound box” task explanation

An experimenter explained the “sound box” punishment task. We developed this paradigm rather than relying on previously established third-party punishment tasks because, at the time of design, we were not aware of any validated online-friendly third-party punishment tasks. We designed the “sound box” task to involve the removal of good sounds and the allocation of bad sounds as punishments for two reasons. First, previous adult work has used “sound blasts” as punishments (Pedersen et al., 2018) and, second, we thought children would intuitively grasp how listening to good sounds would be pleasant and listening to bad sounds would be aversive.

To explain the task, the experimenter told participants that they will prepare a “sound box” for six different main characters (henceforth referred to as *Targets*). To provide an example ([Figure 2a](#)), the experimenter showed participants a “sound box” that belongs to a child named Joan and an “unused sound box” that does not belong to anyone. The “sound box” that belonged to Joan had six good

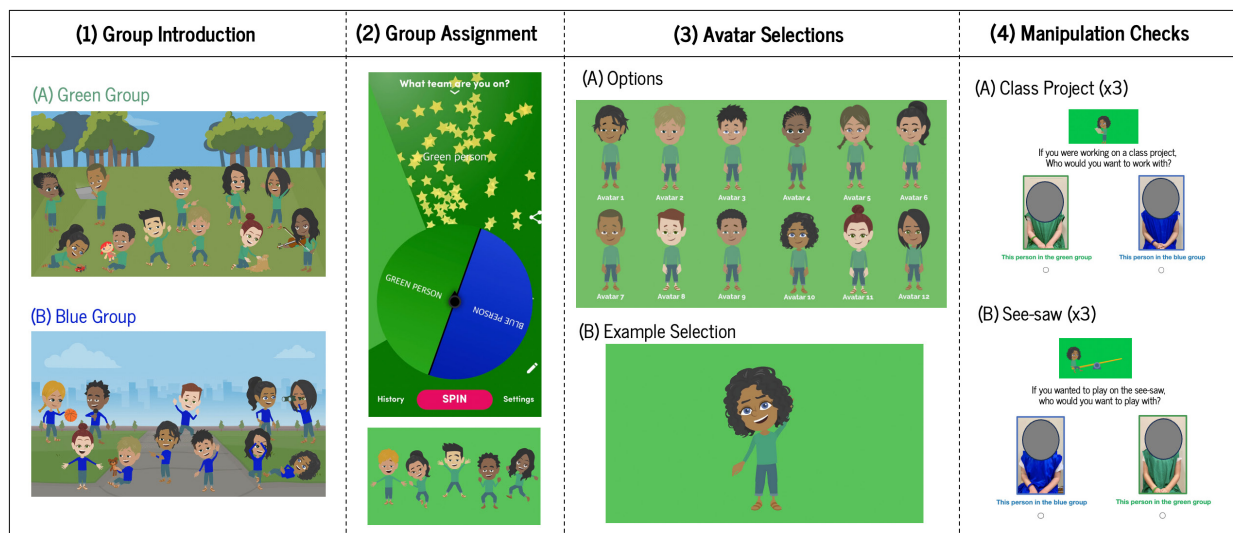


FIGURE 1 Schematic of the group induction procedure. Images of children are blocked for anonymity; participants will see non-blocked versions. Images are copyrighted by and used with permission of VYOND™. VYOND™ is a trademark of GoAnimate, Inc., registered in Argentina, Australia, Brazil, Chile, the European Union, Hong Kong, India, Indonesia, Israel, Japan, Mexico, New Zealand, Norway, OAPI, the Philippines, Russia, Singapore, Switzerland, the United Kingdom, and Vietnam.

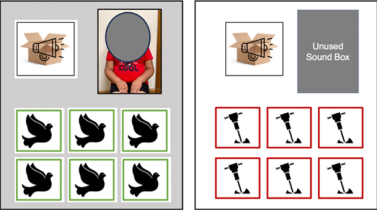
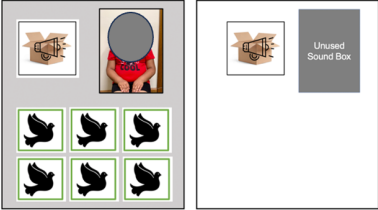
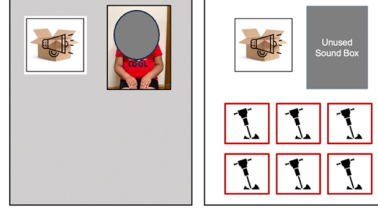
(a) Sound Box Task Explanation	(b) Positive Resource Removal	(c) Negative Resource Allocation
 Let's pretend you're creating a sound box for someone named Joan. You'll be able to see what sounds are in Joan's sound box by looking on this side of the screen in gray. See it has her picture. There is also an unused sound box. You'll be able to see what sounds are in the unused sound box by looking on this side of the screen in white. See it says, "Unused Sound Box". To start, there are six sounds in Joan's sound box, and six sounds in the unused sound box. The sounds in Joan's sound box are good sounds. They're sounds of nature, like birds. All the kids I'll show you today, including Joan, really love to hear the good sounds. They want to hear six good sounds in their sound box and are expecting to hear six good sounds in their sound box—they make the kids really happy. The sounds in the unused sound box are bad sounds. They're sounds of machines, like a jackhammer. All the kids I'll show you today, including Joan, really hate to hear the bad sounds. They do not want to hear any bad sounds in their sound box and are not expecting to hear any bad sounds in their sound box—they make the kids really unhappy.	 Let's talk about Good Sounds. For the Good Sounds, you can leave some or all of them in Joan's sound box, if you want. And, whatever you leave in here, Joan will get to listen to. Or you can take some of them out of this box and put them into this box (<i>pointing toward the unused sound box</i>). And, whatever you move to here, Joan will not get to listen to. And, if you take some good sounds out of this box and put into this one, Joan (and the other kids I'll show you today) will be sad because, remember, they are expecting and wanting to listen to all six good sounds. So, let's make sure you understand: 1) Could you leave all the good sounds in this box (<i>pointing toward Joan's sound box</i>) if you want? Yes or no? That's right; you could leave them all in here if you want 2) Could you move good sounds out of this box (<i>pointing toward the Joan's sound box</i>) and put them into this box (<i>pointing toward the unused sound box</i>) if you want? Yes or no? That's right; you could move 1, 2, 3, 4, 5, or 6	 Let's talk about Bad Sounds. For the Bad Sounds, you can leave some or all of them in the unused sound box, if you want. And, whatever you leave in here, Joan will not have to listen to. Or you can take some of them out of this box and put them into this box (<i>pointing toward Joan's sound box</i>). And, whatever you move to here, Joan will have to listen to. And, if you take some bad sounds out of this box and put into this one, Joan (and the other kids I'll show you today) will be sad because, remember, they are not expecting or wanting to listen to any bad sounds. So, let's make sure you understand: 1) Could you leave all the bad sounds in this box (<i>pointing toward the unused sound box</i>) if you want? Yes or no? That's right; you could leave them all in here if you want 2) Could you move bad sounds out of this box (<i>pointing toward the unused sound box</i>) and put them into this box (<i>pointing toward Joan's sound box</i>) if you want? Yes or no? That's right; you could move 1, 2, 3, 4, 5, or 6

FIGURE 2 Visualization and verbatim scripts for the "sound box" task. (a) Sound Box Explanation, (b) Explanation of "Positive Resource Removal," (c) Explanation of "Negative Resource Allocation." All participants heard the "sound box" task explanation first and also comprehension checks (not depicted here). Whether participants learned about Positive Resource Removal first was counterbalanced across participants. Images of children are blocked for anonymity; participants will see non-blocked versions.

sounds in it (which are valued by the Targets), while the “unused sound box” included six bad sounds (which are disliked by the Targets). The experimenter then asked a series of comprehension checks to confirm children understood the contingencies of the sound boxes. For these comprehension checks (and all others in the study), the experimenter corrected participants if they responded incorrectly up to three times; if they still did not answer correctly after these corrections, we considered this as a comprehension check failure. As described in Participants, these individuals were still included in analyses.

Next, the experimenter explained that participants will make decisions about the good sounds (i.e., Positive Resource Removal) and bad sounds (i.e., Negative Resource Allocation) separately for each Target. For ease of understanding, the experimenter explained the good sounds (i.e., Positive Resource Removal) and bad sounds (i.e., Negative Resource Allocation) separately; we counterbalanced whether the experimenter discussed good or bad sounds first between-subjects and maintained that order throughout the remainder of the experiment.

For *good sounds* (i.e., Positive Resource Removal; exact script in Figure 2b), the experimenter presented participants with an example Target who had a “sound box” filled with six good sounds and an “unused sound box” filled with zero good or bad sounds. For *bad sounds* (i.e., Negative Resource Allocation; exact script in Figure 2c), the experimenter presented participants with an example Target who had a “sound box” filled with zero good or bad sounds and an “unused sound box” filled with six bad sounds. For both good and bad sound decisions, we specified that participants would make those decisions while the experimenter turned off the video to reduce the possibility that participants' decisions are based on reputational concerns (e.g., Jordan & Rand, 2020).

Notably, punishment in the “sound box” task is “free,” like some previous work (Kenward & Öst, 2015) but unlike other work that has employed “costly” tasks (e.g., Marshall et al., 2021). The primary reason for this is that we wanted to make the task simpler given its current length and complexity. Additionally, previous research has found that children are more willing to punish when it is free compared to when it requires sacrifice but not that free situations are particularly likely to elicit punishment relative to costly situations (e.g., McAuliffe et al., 2015). Because of this finding, it is likely that willingness to punish in our task may be heightened relative to previous work on costly punishment. Importantly, though, we did not think this component of the design should influence willingness to punish in *specific* Group Membership or Punishment Type conditions—the key aspect of this study—although see the General Discussion for greater detail on this aspect of the design.

Next, an experimenter explained three final aspects of the “sound box.” First, to ensure that participants believed the Targets would actually listen to the “sound box” (rather than mute the bad sounds), the experimenter

explained how, when they meet with the Target, the experimenter will supervise each Target to ensure each sound in the “sound box” is actually played. Second, an experimenter explained how the Targets will not be able to trace the sounds in the “sound box” back to the participant, reducing the possibility that participants' choices are influenced by fear of retaliation on behalf of the Target or reputational concerns. Third, the experimenter explained how Targets will not receive any information with their “sound box,” so Targets will not know why they have to listen to the “sound box”—Targets will just listen to the sounds without any context. Because of this feature of the “sound box,” there is no way for Targets to learn a lesson about their own behavior based on what is in the “sound box.” This aspect of the design—based on previous research (Marshall et al., 2021)—ensured that punishment satisfied only retributive motives and not consequentialist ones. At this point, we asked a series of comprehension checks to confirm that children understood the contingencies of the experiment. For example, we confirmed that participants remembered that, if participants take away a good sound or give a bad sound, the Targets will be sad. To measure the degree to which participants view these punishments as retributive, the experimenter followed up by asking, “How sad will they be? A tiny bit sad, very sad, or somewhere in between?” (depicted via five increasingly large sad face emojis above a 5-point scale). The experimenter also asked participants to determine whether removing a good sound or giving a bad sound is worse and to what extent (“How much worse? A tiny bit worse, much worse, or somewhere in between?” depicted via five increasingly sized circles above a 5-point scale).

Test trials

Participants then proceeded through six test trials in randomized order that varied on two within-subjects dimensions—Group Membership (ingroup, outgroup, no group; relative to the participant) and Behavior Type (antisocial, neutral) (Figure 3a). Among these six Targets, we paired each boy Target with a girl Target to limit the possible number of Target combinations among the six. We tethered the Group Membership of both Targets within a boy-girl pair to one another and manipulated between-subjects whether the Targets within a pair were portrayed as either ingroup, outgroup, or non-group members; that is, approximately one-third of participants saw Targets within a specific boy-girl pair as ingroup members, another approximate third saw those same Targets within a specific boy-girl pair as outgroup members, and a final approximate third saw the Targets within a specific boy-girl pair as non-group members. For all trials, the Targets were presented on their own (not within a pair, as shown in Figure 2); the “pairing” was only portrayed when introducing all the Targets together (as shown in Figure 3).

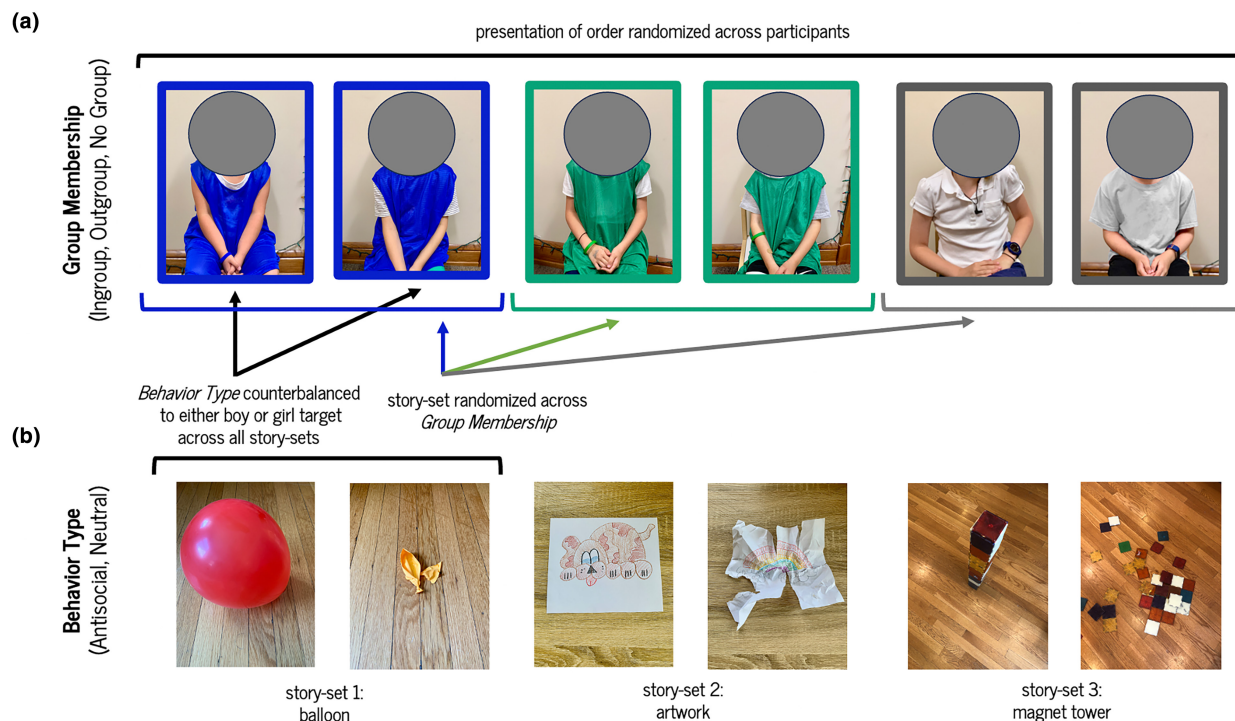


FIGURE 3 Schematic of configuration of six test trials. The six trials varied on two dimensions: (a) Group Membership (ingroup, outgroup, no group; within-subjects) and (b) Behavior Type (antisocial, neutral; within-subjects). Images of children are blocked for anonymity; participants will see non-blocked versions.

One Target within each Group Membership category was described as acting antisocially, whereas the other was described as acting neutrally. Across all trials, either the girls (or the boys) were always described as antisocial (vs. neutral); we counterbalanced whether participants saw girls as antisocial (or neutral). For all trials, we explained how the Target was playing with another child who is a victim (in the antisocial trials); this other person was described as a non-group member across all trial types. We did so because of research finding that children's punishment decisions vary as a function of the social group membership of the victim (e.g., Jordan et al., 2014), and we therefore wanted to hold this aspect of the design constant.

Further, to reduce redundancy and increase believability, we devised three story-sets that involved property destruction (or not; Figure 3b). These story-sets involved destroying (in the antisocial condition) or watching (in the neutral condition) a piece of artwork, a balloon, or a tower made out of magnetic blocks. For example, the experimenter said the following for the artwork story, "Now, I'm going to tell you a story about something that happened one day. This is Jenna; she's a green person. This is Emma; she's not in a group yet. Jenna and Emma were playing together. While they were playing, Emma drew a picture. When they were done playing, Emma asked Jenna to watch her picture. When Emma walked away, Jenna ripped up (watched) Emma's picture" (non-parenthesis indicate script for antisocial condition; parenthesis indicate script for neutral condition). We

randomized which Group Membership set corresponded to which story-set (balloon, artwork, magnet tower).

After presenting the story, an experimenter asked each participant to judge the action: "Do you think this was good or bad?" Depending on their response, we asked a follow-up question: "How good (bad)? A tiny bit good (bad), very good (bad), or somewhere in between?" (depicted via five increasingly sized circles above a 5-point scale). Next, we will ask participants three questions about their emotions in randomized order: anger, sadness, and happiness. For example, the experimenter asked, "Does this make you feel sad (angry) (happy)? Yes or no?". If participants responded "yes," the experimenter followed up by asking participants to rate their degree of sadness, anger, and happiness: "How sad? A tiny bit sad, very sad, or somewhere in between?" This question was accompanied by five sad (angry) (happy) faces in increasing size and corresponding increasing numbers.

Participants then created the "sound box" for the Target in question—either making decisions about good sounds first (Positive Resource Removal) and bad sounds (Negative Resource Allocation) second or vice versa (counterbalanced across participants; consistent within-participants, as described above). Specifically, participants stated their decisions out-loud by saying, "I want to remove [0, 1, 2, 3, 4, 5, 6] good sounds from _____'s sound box" (in the Positive Resource Removal condition) and "I want to add [0, 1, 2, 3, 4, 5, 6] bad sounds to _____'s sound box" (in the Negative Resource Allocation condition).

Finally, because it was very important for participants to remember the retributive nature of punishment in this task, we included a reminder of this after three test trials (halfway through the test trials). In particular, we re-stated the three features of the “sound box” that were explained earlier in the task: (1) Because the experimenter will be present, the Targets will listen to all the sounds in their box, if there are any in it at all; (2) Targets will not know that participants decided the sounds in the “sound box”; and, (3) because Targets do not receive any information with their “sound box,” Targets will not know why they have to listen to the particular sounds (if there are any at all) and, as a result, cannot learn a lesson about their own behavior based on the sounds in the “sound box.”

RESULTS

Our analyses are structured in the following way. First, we confirmed that two of the experimental manipulations were successful: (a) did the minimal group manipulation produce ingroup bias? and (b) did children view Negative Resource Allocation as worse than Positive Resource Allocation? Second, we addressed our primary aims outlined in Table 1. Third, we addressed the exploratory aims outlined in Table 1. Finally, we also conducted additional data analyses (trial order effects, ingroup bias effects, and participants' responses to the “End-of-Experiment” questions), which have been relegated to the [Supporting Information](#) for the sake of brevity.

Did the experimental manipulations work?

First, we examined whether our manipulation successfully induced ingroup bias as would be predicted from previous work (Yang & Dunham, 2019). We found the manipulation did elicit ingroup bias: children's selections on the six manipulation checks significantly differed from chance, $t(147)=7.07, p<.001$; $M=3.82, SD=1.42$ (see Figure S3).

Second, we examined whether children perceived Negative Resource Allocation (giving bad sounds) as worse than Positive Resource Removal (taking away good sounds). In line with our predictions, children rated giving bad sounds ($M=3.84, SD=1.40$) as eliciting greater sadness than taking away good sounds ($M=3.58, SD=1.44$) on a 1–5 scale with 5 indicating the maximum sadness. Indeed, the difference in how children rated Negative Resource Allocation from Positive Resource Removal significantly differed from zero, $t(150)=2.64, p=.009$. We also asked children which was worse, to remove good sounds to give bad sounds, and 79% of children ($SD=0.41$) indicated that giving bad sounds was worse than taking away good ones. This percentage of participants significantly differed from chance, $t(150)=8.63, p<.001$.

In summary, these results indicate that our group manipulation did successfully induce ingroup bias and that children viewed giving bad sounds as a more harmful (and therefore more retributive) punishment.

Primary Aim 1: Do children punish outgroup transgressors more than ingroup transgressors in explicitly retributive contexts?

We first considered whether children punished outgroup members more than ingroup members independent from the type of punishment. To address this question, we built both linear mixed effects models and generalized linear mixed effects models (specifying both a Negative Binomial and Poisson distributions because participants' decisions represent count data) with the relevant predictor variables, notably Group Membership (ingroup, nongroup, outgroup; within-subjects) and Behavior Type (antisocial, neutral; within-subjects). For both sets of models, we included participant ID as a random effect to account for variability between participants. Model 1 was an intercept-only model (with participant ID as a random effect). Model 2 included the main effects of Group Membership and Behavior Type. Model 3 added the two-way interaction between these two predictors. Model comparisons (along with all others conducted here) were calculated via likelihood ratio tests. We examined residual plots (Q-Q plots) to determine whether we should rely on linear mixed effects models or generalized linear mixed effects models (with either a Negative Binomial or a Poisson distribution). The residual plots indicated that all three types of models are appropriate for the given data, and—because the linear mixed effects model residual plot looked somewhat better than both the Negative Binomial and the Poisson plots—we opted for linear mixed-effects models here.

Model comparisons revealed that Model 2 (the main effect model) exhibited improved model fit, $\chi^2(3)=618.30, p<.001$, over Model 1 (the intercept-only model). Aligning with other work finding that children around age six punish antisocial actions (e.g., Marshall & McAuliffe, 2022), this model revealed a main effect of Behavior Type, $\chi^2(1)=745.30, p<.001$: participants punished antisocial actions ($M=2.43; SE=.12$) more often than neutral ones ($M=.55, SE=.12$). We did not find a main effect of Group Membership, $\chi^2(2)=2.63, p=.268$: participants punished nongroup members the most ($M=1.54, SE=.12$) followed by outgroup members ($M=1.52, SE=.12$), and then ingroup members ($M=1.41, SE=.12$), although none of these groups differed from one another. Contrary to our predictions, Model 3 (two-way interaction model) did not exhibit improved model fit beyond Model 2, $\chi^2(2)=2.58, p=.275$. Accordingly, the Group Membership $\times$ Behavior Type interaction was not significant, $\chi^2(2)=2.57, p=.276$. As illustrated in Figure 4, although participants punished more when confronted with an antisocial action

relative to a neutral one, participants did not differ in the extent to which they were willing to punish ingroup, nongroup, and outgroup members in the antisocial cases.

Primary Aim 2: Do children engage in even greater outgroup punishment in contexts where punishment is perceived as especially retributive (i.e., Negative Resource Allocation)?

We next considered the possibility that Group Membership only affects punishment behavior within a particular Punishment Type (Positive Resource Removal, Negative Resource Allocation; within-subjects). To investigate this possibility, we compared similar models as specified for Primary Aim 1 but also included Punishment Type as a predictor variable, rendering four different models. In particular, Model 1 was the intercept-only model. Model 2 included the main effects of the three predictor variables (Group Membership, Behavior Type, and Punishment Type). Model 3 included the two-way interactions between these three variables. Model 4 included the three-way interactions between these three variables.

Model 4 did not exhibit improved model fit beyond the other models, $\chi^2(2)=0.75$, $p=.686$, indicating that Group

Membership did not affect participants' willingness to punish for either type of punishment. As illustrated by Figure 5, participants engaged in similar sorts of punishment regardless of the group or punishment context.

Exploratory Aim 1: In what ways are the hypothesized patterns dependent on age?

It is possible that the lack of significant predicted interactions for both Primary Aim 1 and 2 is a result of age-related effects.

For Exploratory Aim 1, we constructed the same sets of models for model comparison as was the case for Primary Aim 1 but added an additional predictor of Age (continuous). Specifically, these models included Group Membership, Behavior Type, and Age. The model that included this three-way interaction did not exhibit improved model fit, $\chi^2(2)=0.10$, $p=.951$, over a model that did not include that interaction; correspondingly, the Group Membership $\times$ Behavior Type $\times$ Age interaction was not significant, $\chi^2(2)=0.10$, $p=.951$. The model that only included the two-way interactions exhibited the best model fit, $\chi^2(5)=25.35$, $p<.001$, relative to the model that did not include two-way interactions.

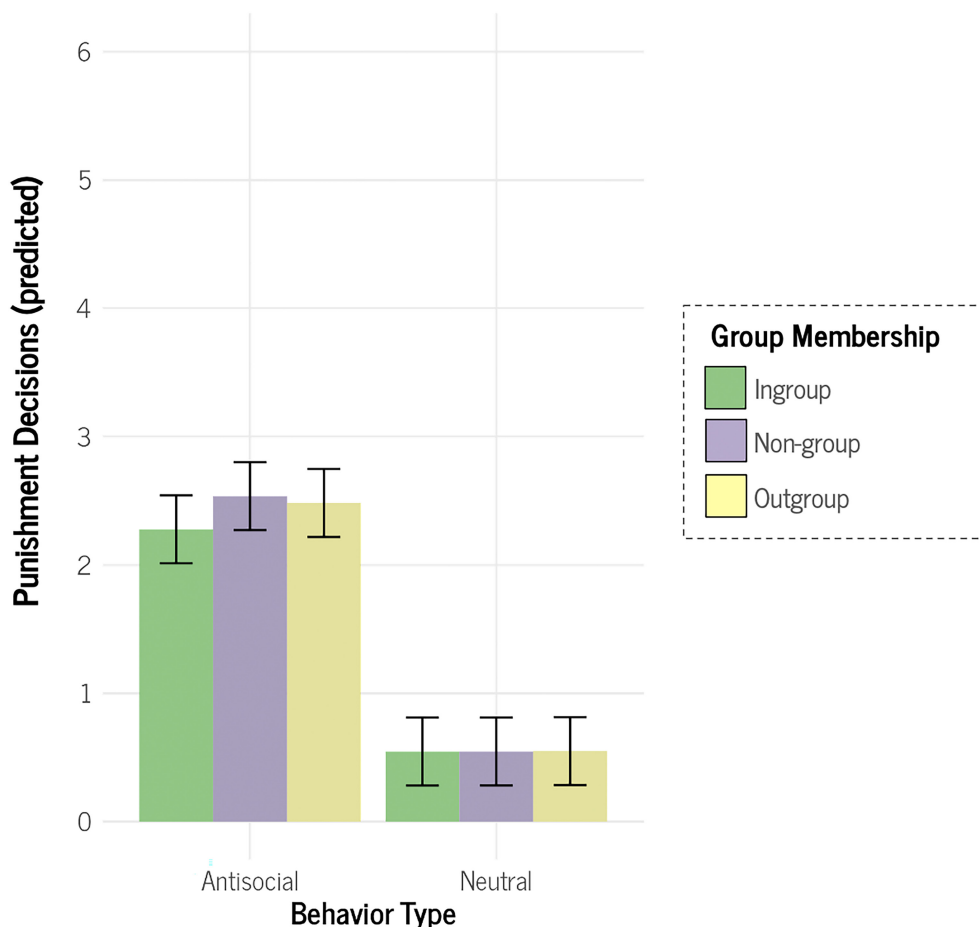


FIGURE 4 Punishment decisions as a function of Group Membership and Behavior Type. Error bars represent 95% confidence intervals.

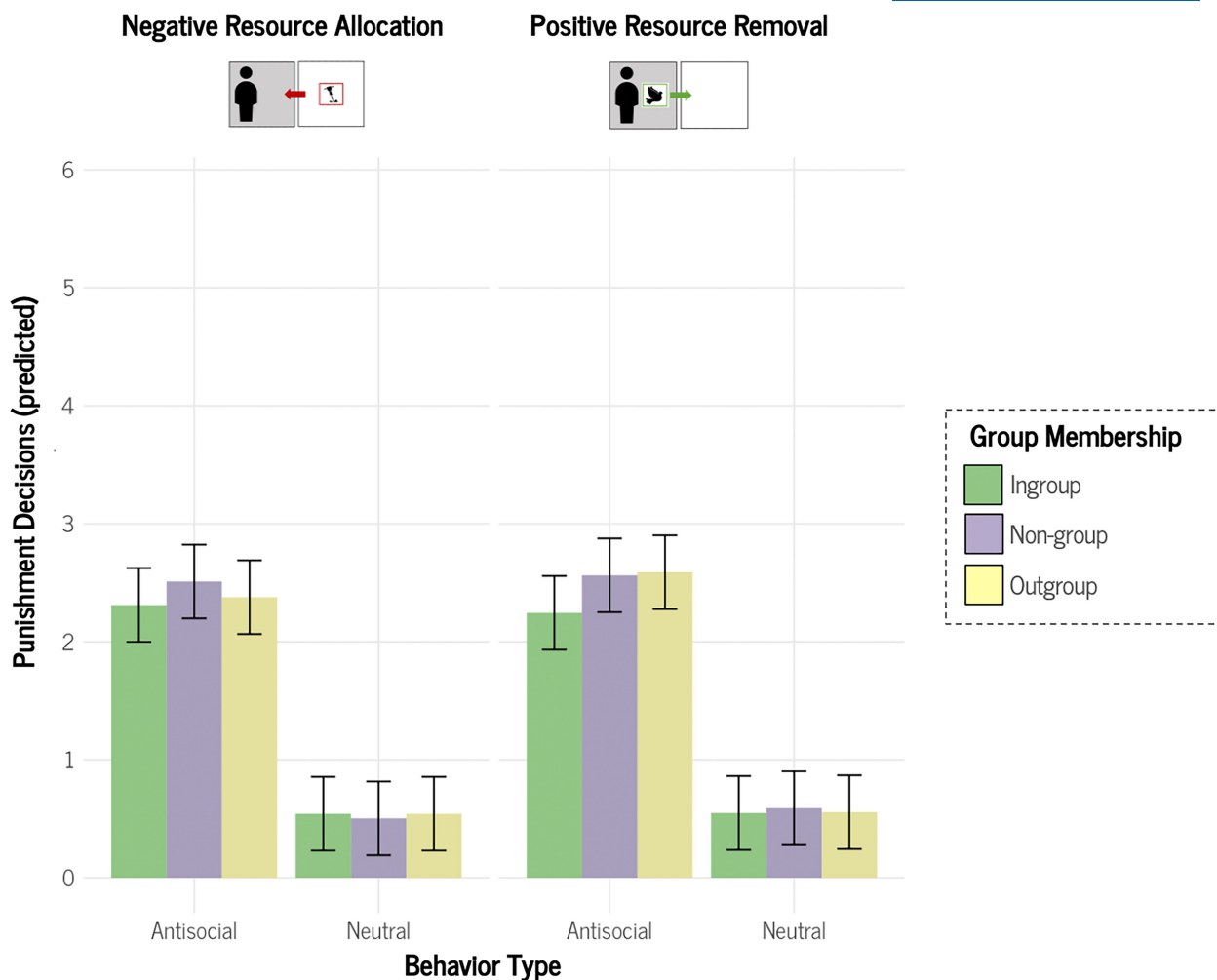


FIGURE 5 Punishment decisions as a function of Group Membership, Behavior Type, and Punishment Type. Error bars represent 95% confidence intervals.

Looking at the model that included the two-way interactions revealed a two-way interaction between Behavior Type and Age, $\chi^2(1)=21.03$, $p<.001$; the other two-way interactions were not significant, $ps>.273$. As found in previous research (Lee & Warneken, 2022a), this interaction revealed that younger children were less willing to engage in third-party punishment compared to older children, and this pattern appears to result from age-related decrease in the willingness to punish neutral actions, $B=-.29$, $p=.010$, but not antisocial actions, $B=-.01$, $p=.920$ (Figure S4).

Next, we constructed the same sets of models for model comparison as was the case for Primary Aim 2 but added in an additional predictor of Age (continuous). In particular, the models included Group Membership, Behavior Type, Punishment Type, and Age. The model that included this four-way interaction did not exhibit improved model fit, $\chi^2(2)=0.40$, $p=.818$, over a model that did not include that interaction; correspondingly, the Group Membership $\times$ Behavior Type $\times$ Punishment Type $\times$ Age interaction was not significant, $\chi^2(2)=0.40$, $p=.820$.

Exploratory Aim 2: How does group context change evaluations of and emotional responses to antisocial behavior? Do potential differences predict variation in punishment behavior?

We next considered children's evaluations and emotional responses. First, we examined whether Group Membership and Behavior Type predict children's evaluations of and emotional responses (sadness, anger, happiness) to the Targets. To address such a possibility, we built linear mixed-effects models: Model 1 only included participant ID as a random effect. Model 2 included the main effects of Group Membership and Behavior Type. Model 3 added the two-way interaction among these variables.

For evaluations, the planned models would not converge, likely because of ceiling- and floor-like responses. As a result, we compared an intercept-only model to a model with a main effect of Group Membership for both antisocial and neutral actions. For both antisocial and neutral actions, the model including the main effect of Group

Membership did not exhibit improved model fit beyond the intercept-only model (Antisocial: $\chi^2(2)=5.51$, $p=.064$; Neutral: $\chi^2(2)=1.58$, $p=.454$). Generally, participants described Antisocial actions as “bad” and neutral actions as “good” and such evaluations did not vary depending on who committed the act (Figure S5). In additional exploratory analyses, we examined whether Age (continuous) interacted with Group Membership for both Antisocial and Neutral Actions, and it did not for either, $ps>.228$.

For all three emotions, we found that Model 2 (the main effects model) exhibited improved model fit (sadness: $\chi^2(3)=482.57$, $p<.001$; anger: $\chi^2(3)=376.80$, $p<.001$; happiness: $\chi^2(3)=769.15$, $p<.001$) over Model 1 (the intercept-only model). We found a main effect of Behavior Type for all emotions (sadness: $\chi^2(1)=672.77$, $p<.001$; anger: $\chi^2(1)=485.26$, $p<.001$; happiness: $\chi^2(1)=1326.61$, $p<.001$) such that participants indicated greater sadness and anger in response to antisocial Targets relative to neutral ones and indicated greater happiness in response to neutral Targets relative to antisocial ones. Further, we did not find a main effect of Group Membership ($ps>.490$) such that participants did not indicate differing levels of sadness, anger, or happiness in response to ingroup, outgroup, or nongroup Targets. Furthermore, we did not find an interaction between Behavior Type and Group Membership for any of the emotions ($ps>.695$; Figure 6), meaning that children's emotional responses to antisocial actions did not differ depending on the group membership of the

transgressor. We also examined whether the lack of significant interactions was due to age-related interactions, and this was not the case; Age did not interact with Group Membership and Behavior Type for any of the emotions ($ps>.546$). Collectively, these data suggest that participants felt negative emotion in response to antisocial actions and positive emotion in response to neutral ones, and whether participants experienced such emotions did not depend on whether children witnessed antisocial behaviors committed by ingroup, outgroup, or non-group members.

Next, we considered participants' evaluations of and emotional responses to Targets as predictors of punishment behavior. In particular, we reasoned that anger may uniquely predict Negative Resource Allocation (compared to Positive Resource Removal) punishment in response to antisocial outgroup members. To address such a possibility, we built linear mixed-effects models: Model 1 was an intercept-only model (with just participant ID as a random effect). Model 2 included Group Membership, Behavior Type, Punishment Type, and Evaluation (continuous) OR Emotion (continuous) as main effects. Model 3 included the two-way interactions among these variables. Model 4 included the three-way interactions among these variables. Model 5 included the four-way interaction among these variables. Note that, for brevity's sake, these analyses did not include age; for these analyses, please see the [Supporting Information](#).

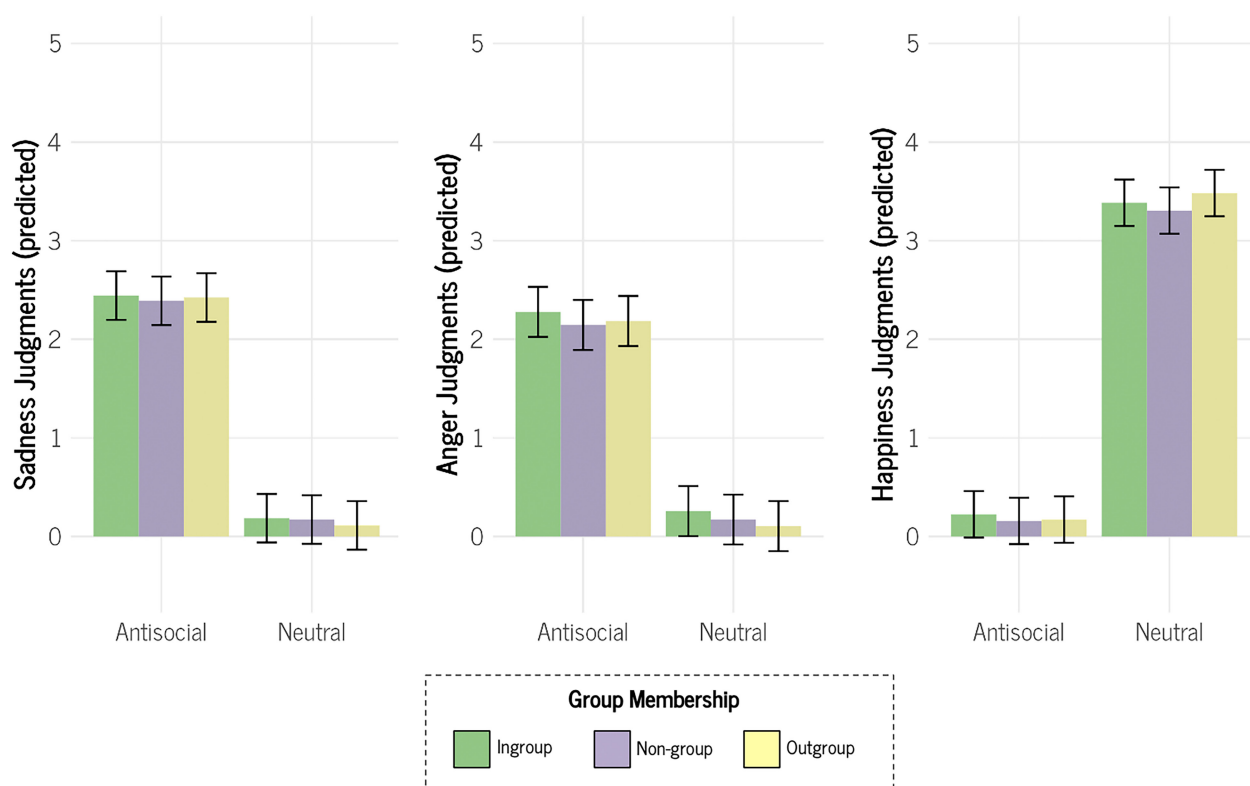


FIGURE 6 Emotional responses as a function of Group Membership and Behavior Type. Error bars represent 95% confidence intervals.

For evaluations, we did not find any interactions among the predictors of interest such that Model 2 (the main effect model) exhibited the best model fit, $\chi^2(5)=693.74$, $p<.001$, compared to Model 1 (the intercept-only model). Looking at Model 2 revealed a main effect of Evaluations, $\chi^2(1)=76.33$, $p<.001$, such that the more participants viewed an action as bad positively related to the degree to which they were willing to punish, $B=.19$, $SE=.02$, $p<.001$. For a visualization, see Figure S6. In other words, participants who evaluated an action more negatively were more likely to punish such behavior regardless of (a) whether that behavior was actually antisocial or neutral (Behavior Type), (b) whether the behavior was committed by an ingroup or outgroup member (Group Membership), or (c) whether punishment required the removal of positive resources of the allocation of negative ones (Punishment Type).

For emotions, we did not find any interaction effects among any of the relevant predictor variables: Model 2 (the main effects model) exhibited the best model fit (sadness: $\chi^2(5)=665.48$, $p<.001$; anger: $\chi^2(5)=738.94$, $p<.001$; happiness: $\chi^2(5)=649.33$, $p<.001$) compared to Model 1 (the intercept-only model). Looking at Model 2, we find a main effect of each emotion (sadness: $\chi^2(1)=47.08$, $p<.001$, $B=.19$, $SE=.03$; anger: $\chi^2(1)=123.76$, $p<.001$, $B=.29$, $SE=.03$; happiness: $\chi^2(1)=30.83$, $p<.001$, $B=-.15$, $SE=.03$) predicting decision. These data suggest that the experience of sadness and anger (in addition to the lack of happiness) promotes punitive behavior regardless of all the other potential factors included in the experimental setup, including Group Membership, Behavior Type, and Punishment Type.

GENERAL DISCUSSION

The present work sought to examine how retributive motives shape the expression of third-party punishment in intergroup contexts. In particular, we hypothesized that—by explicitly isolating and manipulating the retributive aspect of punishment—we would replicate and extend previous results (e.g., Jordan et al., 2014) finding that children direct third-party punishment toward outgroup transgressors relative to ingroup ones. In doing so, we predicted that the present work would provide clarity regarding why children direct punishment toward outgroup transgressors relative to ingroup transgressors, even though previous theorizing has argued that we should expect the opposite (the “norm enforcement hypothesis”; McAuliffe & Dunham, 2016). The present data did not confirm our prediction. Instead, we found that children, regardless of age, similarly directed retributive punishment toward ingroup, outgroup, and nongroup transgressors and that the degree to which punishment was perceived

as retributive did not affect punishment behavior. Importantly, though, children did exhibit ingroup bias and also distinguished between the retributive value (i.e., the harmfulness) of the two types of punishment. On the whole, these findings suggest that a desire to inflict retribution is not specifically tethered to punishing outgroups, at least within the presently used paradigm.

The present results raise many questions. For one, why do children similarly punish ingroup, outgroup, and nongroup members? On the one hand, it is plausible that the group manipulation was not particularly effective. In favor of this possibility, the manipulation did occur very early in the task and participants may have been less invested in the group nature of the task by the test trials. On the other hand, we did design the experiment to minimize this possibility; we reminded participants of their group membership before the test trials, explicitly mentioned the group membership of the transgressor and victims on each trial, placed an image of the participant's avatar on each test trial, and found that participants generally exhibited ingroup bias on the manipulation check items. In line with previous work (Yang & Dunham, 2019), these particular results suggest that the manipulation was effective. Nonetheless, it is worth mentioning that the ingroup bias found in that work was quite a bit higher ($M \sim .80$) than documented here ($M \sim .64$) on analogous but not identical items. Relatedly, it is possible that the minimal group manipulation was perhaps less potent in the online context, meaning that ingroup bias on a judgment item was not powerful enough to translate into differential punitive behavior. To this point, children in an online task have very little context for why they are assigned to a group and have little expectation of ever interacting with anyone in their group. It is challenging to know whether testing children in an online environment may explain the lack of intergroup bias effects, but work by Leshin et al. (2022) counters this possibility given that they find intergroup bias in children's third-party punishment in online settings.

Beyond the group manipulation, punishment in the present task did not require a personal cost. We opted to make punishment-free because previous research has found that, although children punish more when it is free compared to when it is costly, neither free nor costly situations are uniquely likely to promote third-party punishment of antisocial behavior (McAuliffe et al., 2015). By making punishment costless, we reasoned that children's punishment rates may be higher in general but that the free versus costly nature of punishment would likely not interact with intergroup context. Against this possibility, Leshin et al. (2022) found that, although children's costly third-party punishment and non-costly third-party punishment were highly correlated, intergroup bias effects in punishment as a function of parents' political ideology emerged for costly punishment but not non-costly

punishment. It is therefore possible that had punishment been costly, children may have been more discerning in their investment of when and whom to punish, which could have then resulted in more stark intergroup bias effects. That being said, it is also possible that the opposite is true: perhaps the children who engage in costly punishment are particularly committed to punishment and are correspondingly less likely to consider outside factors when deciding whether to punish. Regardless, it is challenging to draw any firm conclusions about the role of cost in shaping intergroup bias in punishment because much of the work to date yields mixed results and rely on costly paradigms. In particular, one study finds greater ingroup punishment (Gonzalez-Gadea et al., 2022), another finds greater outgroup punishment (Jordan et al., 2014), and a third finds no difference (among older children in particular; Gummerum et al., 2009; Yudkin et al., 2020).

Another potential explanation for the lack of intergroup effects relates to antisocial action itself. Specifically, perceived severity of an antisocial action may moderate people's propensity to punish ingroups versus outgroups. For instance, some may reason that certain antisocial actions are so severe that anyone who commits them—regardless of group membership—deserves to receive punishment. In support of this, children endorse similar punishment for agents, regardless of group membership, who engage in severe antisocial actions (Schuhmacher & Kärtner, 2019) and judge ingroup and outgroup members as similarly wrong for engaging in more severe antisocial actions, such as hitting (Mulvey, 2016; Schuhmacher & Kärtner, 2019). Relatedly, work has found attenuated intergroup bias in endorsement of punishment when the antisocial action is intentional compared to accidental (D'Esterre et al., 2022). Meanwhile, work focused on fairness norm violations, a relatively mild offense (Cameron et al., 2022), has revealed ingroup bias in punishment (e.g., Gonzalez-Gadea et al., 2022; Jordan et al., 2014). Thus, in the context of our task, it is possible that the antisocial actions presented to children were too severe to generate group bias effects in punishment that may have revealed themselves in the context of less severe antisocial actions. However, evidence against this explanation comes from other work that has relied on more severe antisocial actions and *does* find intergroup effects. For example, 4-year-olds are more inclined to judge that an outgroup member who hits or kicks someone should receive punishment compared to an ingroup member (Chapman et al., 2020). Further, children between the ages of three and eight exhibit intergroup bias in punishment in response to property destruction, although the direction of this bias depends on certain factors (such as the political ideology of participants' parents; Leshin et al., 2022; Yudkin et al., 2020).

A final explanation relates to the options available to participants. It is possible that children feel a desire to respond to all antisocial actions but that the only option

available to them is punishment. Coupled with a strong desire to respond to such severe actions, children may punish ingroup and outgroup transgressors alike simply because they do not want to ignore either ingroup or outgroup antisocial actions, even though they may prefer to respond in different ways. This hypothesis makes the prediction that, if additional options are made available (such as compensation; FeldmanHall et al., 2014), participants may exhibit differential strategies in how they respond to ingroup versus outgroup transgressors. For example, children may be more inclined to punish outgroup transgressors but more inclined to compensate the victims of ingroup transgressors. Even still, these sorts of arguments were in part the reason why we provided participants not only with two different forms of punishment but also allowed them to punish continuously; despite this element of the design, it is important to acknowledge that the only options available to participants were punitive, not compensatory or rehabilitative. These broad ideas relate to a debate within social and developmental psychological research regarding whether rates of punishment are artificially inflated in studies because participants are typically only provided with the option to punish transgressors or to do nothing in response to antisocial actions (e.g., Pedersen et al., 2018). Perhaps by introducing additional options to respond to wrongdoing, future research can better clarify intergroup bias in third-party punishment.

Beyond addressing the lack of intergroup bias in punishment, there is also a question about why the type of punishment did not affect bias in punishment even though participants acknowledge that allocating negative sounds was worse than removing good sounds. Here, too, we have generated many post-hoc explanations for this lack of interaction, although we clearly predicted that there would be enhanced intergroup bias in the context of Negative Resource Allocation compared to Positive Resource Removal.

One explanation that concerned us was that participants were misunderstanding the procedure. After testing a few participants, it came to our attention that some participants may erroneously think that the removal of a good sound necessitates the addition of a bad sound, rendering it impossible to detect an effect of the type of punishment. The additional questions we asked participants regarding their beliefs about the nature of the sound boxes remedied this concern. Children largely thought that they could make different decisions about the good and bad sounds and also thought that there could be more than six sounds in a box. When only examining children who expressed a proper understanding of the procedure, we also did not find the predicted interaction effects.

Instead, the present results have also required us to reconsider our predictions generally. Although we clearly expected the present study to elicit intergroup bias in punishment, it seems like none of the data

aligns with that possibility. In light of this, it is possible that retributive punishment contexts, rather than promoting outgroup punishment, elicit a willingness to punish that is relatively immune to social context. This possibility is consistent with philosophical discussions of retributive punishment that describe punishment as an obligatory response to wrongdoing that rectifies past injustices (Cottingham, 1979). From these sorts of perspectives, the key factor in determining punishment is the severity of the offense, not the person committing it. Although the present data cannot speak to whether children are explicitly thinking of punishment in this way, their behavior is certainly more consistent with the notion that some children engage in retributive punishment in response to wrongdoing and that willingness is relatively insensitive to surrounding circumstances. In support of this possibility, other work has found that children, especially at a younger age, endorse the notion that individuals should punish one another regardless of social context (Marshall et al., 2020) and also are willing to punish even when doing so does not stand to teach the transgressor a lesson or confer personal benefits (Lee et al., 2024; Marshall et al., 2021; Twardawski & Hilbig, 2020).

This line of thinking relates to recent findings that point toward the possibility that retributive punishment has been poorly misunderstood in psychology research (Gerber & Jackson, 2013). On the one hand, retributive punishment can be thought of as a malicious desire to inflict harm on transgressors—the sort of conceptualization we had taken on here initially. On the other hand, retributive punishment can be thought of as a duty that is required in response to wrongdoing that is motivated by the obligation itself, not by the desire to see harm inflicted on a transgressor (even though the inflicted harm is a necessary byproduct of the obligation)—a view that we had not taken on here but is more consistent with the resulting data. Further research is required to better understand the roots of retributive behavior and such work can benefit our understanding of intergroup bias in cooperative behavior.

Beyond the two key research questions that motivated the present work, the current research also provided some additional insight into the development of third-party punishment (Marshall & McAuliffe, 2022). First, we find age effects with respect to third-party punishment. Although children generally punish antisocial actions, younger children are more inclined to punish neutral actions compared to older children. This effect has been found in other work as well (e.g., Lee & Warneken, 2022a, 2022b). Why younger children punish neutral actions is not especially clear, but it is possible that they have a bias against doing nothing in situations and prefer to move the sounds around regardless of the antisociality of the situation.

Second (and relatedly), we find that many interesting correlations between children's evaluations of and

emotional responses to the presented actions and their willingness to punish. Independent of whether the action was antisocial or neutral, the extent to which children viewed an action as negative and felt increased negative emotions (or decreased positive emotions) predicted their willingness to punish. These findings are important in light of previous results that have found significant relationships between emotional responses and punishment behavior in adolescents (Gummerum et al., 2020, 2022) and adults (Heffner & FeldmanHall, 2022) but not in children (Marshall et al., 2021). Because the present study only tests children, we can examine whether emotions play a more powerful role in shaping punitive behavior for adolescents and adults compared to children, although future research should certainly examine this possibility.

We will end with a few final limitations of the present work. First, in the “Introduction,” section, we discussed how intergroup context has the potential to intersect with punitive motives in two different ways. In particular, we argued that retributive desires may promote outgroup punishment, while consequentialist desires may promote ingroup punishment. Here we focused only on the first component because we wanted to clarify the motives underlying outgroup punishment and correspondingly isolated the retributive value of punishment in our studies by specifying that punishment cannot possibly teach a lesson. Even though we do not find evidence that retributive contexts promote outgroup punishment does not mean that consequentialist contexts do not amplify ingroup punishment. Future research could address this precise question by manipulating the consequentialist value of punishment and see whether ingroup punishment is enhanced in expressly consequentialist contexts.

Second, it is important to acknowledge that the present task is limited in its ecological validity. For one, children in the real world are rarely (if ever) put in the position to punish their peers (see Guala, 2012 for a discussion of this point in adult work). As a result, it is unclear whether children's willingness to punish in the present tasks (and other related third-party punishment tasks) actually corresponds with their punishment behavior in the real world. Indeed, whether children's more spontaneous punishment would reflect intergroup bias is a question for future work. Second, the present study extrapolates broader findings from minimal groups that may have little bearing on the real world. Future work should certainly consider the extent to which the questions at the core of this manuscript apply to real-world groups, especially given pervasive disparities in punishment on the basis of race in everyday life (e.g., Okonofua & Eberhardt, 2015).

On the whole, the present work sought to clarify previous mixed results regarding the intergroup bias in punishment. In particular, we aimed to show that a desire to punish outgroup transgressors may be more closely tethered to a retributive desire to inflict suffering in the

absence of societal benefits. Contrary to this hypothesis, the present results did not confirm this prediction and accordingly stand to muddy already complicated waters even further. Nonetheless, the present results undoubtedly contribute to a growing body of research concerned with understanding how intergroup context influences cooperative behavior (Lazić et al., 2021) and raise many testable questions regarding how group contexts shape third-party punishment.

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DATA AVAILABILITY STATEMENT

The data, code, and materials necessary to reproduce the analyses presented here are publicly accessible <https://osf.io/6uxpe/>. The analyses were pre-registered: <https://doi.org/10.17605/OSF.IO/ZGTNV>.

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SUPPORTING INFORMATION

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